

image of the country in terms of its capability to manufacture LED lights for the world. There are many unorganised players who import unsafe LED light components just because they are cheap. The threat to one's safety is also a concern because of the same.

It is to be noted here that India imported lamps and lightning fittings amounting to a value of over US\$436 million in FY 2020. India, as per Energy Efficiency Services Limited, is the second largest LED lights market by value.

As a matter of fact, the government of India has recently introduced mandatory testing of LED lights imported from outside India. The customs department of the country has been given a free hand to randomly pick up samples of LED lights and get them tested at labs accredited by the authorities.

Interestingly, China is among the world's largest exporters of LED lights and raw materials required for making LED lights. During 2014-15, India imported LED lighting products under HS code 9405 worth US\$ 222.3 million. China was the largest supplier with a share of 84.76 per cent, followed by Republic of Korea with 5.43 per cent and Hong Kong with 2.4 per cent.

Dumping ground to global manufacturing hub

The two questions that need to be answered in order to make India a hub of LED lights manufacture revolve around dumping of components and finished products in India by foreign players and our dependence on imports. Raja explains that if we keep focusing on importing certain components required for manufacturing LED lights, we might not be able to focus on developing an ecosystem for manufacturing these components in the country.

"If we keep importing components into India, we might not feel it to be important for developing these within the country. The government also needs to start by putting up certain entry barriers in forms of duties and legislations to make sure that production of components required for LED lights is promoted," says Mukherjee.

The volumes of LED lights that are being assembled in India might not be



perfect for setting up the component ecosystem here. Arun explains that if India starts getting export orders for LED lights, and the conventional lamps still being produced in India are replaced with LED lights, the volumes will become good enough for component makers to set shop in the country.

"If volume comes, people will definitely invest in manufacturing components for LED lights. Once the volume comes India can easily become the export zone of LED lights," feels Kumar.

The availability of cheaper products, as pointed out by Parag, is a challenge that every business faces. Parag explains, the best way to address this challenge is by adding value for the end-consumer. Investing in R&D and design, as per him, is the best way to move towards the direction of making India an export hub for LED lights.

"The modern consumer, whether from India or from abroad, is a very smart consumer. If you can add value to a product, the modern consumer will value it. I am very happy to say that a lot of manufacturers in India have started focusing and investing on R&D and design to make the quality better," says Bhatnager.

Betting upon smart LED lights

While there are a lot of interesting verticals coming up in the LED lights arena, the one that India can possibly lead in is the smart LED lighting. The trend is not only picking up pace in

India, it is also picking up pace in a lot of developed countries. The market for mood lighting, voice-activated lighting, automatic lighting, use of smart LED bulbs in vehicles, and LiFi have started to show promising growth.

India, as per Kumar, should focus on this vertical given the excellent work IoT developers in the country have shown. India's capability to design IoT solutions around LED lights comparatively economically than a lot of other countries is a big strength. He explains that betting big on this vertical can help the country appear on the globe as the hotspot of manufacturing smart LED lights.

The next big opportunity in terms of LED lights will come from rural India and small towns in the country. Parag explains that these areas have not yet witnessed the advantages in terms of cost savings that LED lights offer.

Further, there are many countries in the world which can be a great market for LED lights manufactured in India. The neighbouring continent of Africa is a big example of how big the opportunity is on the global level. India already exports tons of generic medicine, mobile handsets, and automobiles to the many nations in Africa; LED lights can be an addition to the list.

Mukherjee explains that facade lighting is also a big opportunity. The government is trying to capitalise on night tourism, and facade lighting has a huge role to play in the same, according to him. **EFY**